

Experience

- 2006-10-30 – **System Architect (Distinguished Engineer)**, *Nokia Corp (San Jose, USA)*
Present
- Developed (white-box) requirements for various generations of Infinera ASICs in these areas: G.709 OTN tributary interfaces, FlexE, FlexO, Infinera Proprietary Signals/Formats (used on the line-side interfaces), Overhead processing, Alarm Propagation. Supported the ASIC development team through all stages of development to ensure full compliance to applicable standards.
 - Developed requirements for Layer 1 encryption (AES-GCM, AES-CTR modes) in INFN ASICs: at the ODU layer (in mapper ASIC) & bulk encryption (DSP ASIC). These requirements addressed the following aspects: algorithmic compliance, support for minimizing the cryptographic boundary (for FIPS 140-2 validation).
 - Specified the architecture for supporting protection switching schemes (e.g., 1+1 SNCP, Y-cable) in Infinera transponders and muxponders. This involved the timely reporting of defects to the protection FSMs, FPGA assist to minimize the switching time.
 - Developed requirements for FEC Support for OC3/1GE Optical Supervisory Channels
 - Developed the End-2-End mechanism for Shared Mesh Protection (SMP) for ODU connections. This mechanism has been presented in various contributions to ITU-T SG15 Q9 (OTN Equipment). This work contributed to the ITU-T recommendation G.783.3.
 - Defined the mechanism to support hitless ODUFlex resizing (specific to DTN-X networks). Working with ASIC & Software teams to get the mechanism implemented.
 - Developed requirements related to control channel realization (e.g. trib & line side GCC channels) for several generations of Infinera Network Elements.
 - Developed DTN-X Network Architecture Specification. This specification takes a top-down view of DTN-X networks, and contains an in-depth coverage of these aspects: Layers (i.e. Signals, Formats), Adaptations, OTN atomic function realization, and Defect propagation. This document is used as the basis to derive requirements for the ASICs/FPGAs.
 - Developed DTN Network Element requirements for these areas: GMPLS, L1VPN, Datapath recovery (protection, and restoration).
- 2005-05-10 – **System Architect**, *Fujitsu Network Communications (Richardson, USA)*
2006-10-01
- Participated in the architecture definition of the FW9500 hybrid platform that supported TDM & Packet services (e.g. Point-to-Point Ethernet, VPLS) based on MPLS Pseudowires
 - Developed System Level Requirements for VPLS/H-VPLS, Control Plane High-Availability.
- 2003-12-01 – **Contractor (SW development)**, *Nortel (Richardson, USA)*
2005-05-01
- RSVP-TE for GGSN node: Wrote Functional Spec, Detailed Design & Designer Test plan. Worked on design changes support for a redundant control processor.
 - BGP/MPLS VPNs for GGSN node: Wrote the Designer Test plan, Added support for MPLS & VRF historical statistics.
- 2003-10-01 – **Contractor (SW development)**, *Avici (Billerica, USA)*
2003-11-20
- Resolved software issues related to: PPP, SONET Link Aggregation, MPLS LSP Protection (Fast Reroute).
- 2002-12-24 – **Contractor (Systems Arch)**, *Ciena (San Jose, USA)*
2003-05-31
- Specified the functional requirements for the CoreDirector features: Connection Test Access, Connection Loopback, OSI/DCC, 1GE/10GE tunneling with SONET/SDH VCAT/LCAS).
- 1998-05-08 – **Systems Architect**, *Fujitsu Network Communications (Richardson, USA)*
2002-08-28
- Part of the team tasked with developing the architecture for the first TDM/Packet Hybrid. Primarily responsible for Layer 2 and Layer 3 VPN support over MPLS networks.

- Investigated protection of Point-to-Multipoint (P2MP, or multicast) MPLS LSPs, for the purposes of offering a fully protected TLS service.
- Specified the optical UNI (O-UNI) interface between an MPLS LER (Label Edge Router), and the Optical Network Element (ONE).
- Specification of Network Element Management Information models, and protocol specific realizations (in terms of TL-1 and SNMP)

1984-06-01 – **Member of Scientific Staff, Nortel (Ottawa, Canada)**

- 1998-05-01
- Part of a team defining the End-to-end architecture of a system offering Broadband access.
 - Developed various features for the HFC Cable Modem. E.g. Lightweight ATM signaling protocol, IP NAT (to support multiple attached PCs), Software Upgrade
 - Designed & Developed software for the following layers in the OSI Stack: LAPD (with QoS), Layer 3 (CLNP, IS-IS, ES-IS), and Layer 5 (Session Layer). Implemented performance enhancements at Layers 2 through 4 of the OSI stack
 - Implemented various STREAMs modules (e.g. TPI, Line Discipline Module, Serial Drivers)
 - Implemented a bridge for interconnecting IBM Token Ring LANs via a Frame Relay WAN
 - Analysis, Design, and Implementation of transparent access to (XMS based) remote file servers (via X.25 networks), file transfer utilities.

1981-04-01 – **SW Designer, Micom (Montreal, Canada)**

- 1984-06-01
- Designed and Implemented software for data communication protocols (e.g. BISYNC, SDLC, ISO Transport Layer 4/0, Session/Transport layers for Teletext service)

Education

1987-09-01 – **Ph.D. (Electrical Engineering), Carleton University (Ottawa, Canada),**

- 1991-01-31
- Thesis: Learning Algorithms for Data Storage & Retrieval
 - Supervisor: Prof. B. John Oommen

1979-01-01 – **M. Sc. (Computer Science), McGill University (Montreal, Canada),**

- 1981-04-01
- Thesis: A Communications Controller for Small-Computer Networks
 - Supervisor: Prof. Kasi Ananthanarayanan

1973-07-01 – **B. Tech. (Electronics and Electrical Communication), Indian Institute of**

- 1978-05-01
- Technology (Kharagpur, India),*
- Undergraduate Project: Symbolic Circuit Analysis

Awards

- 2024-09-01 – Infinera Spectrum Award. Part of the team that solved a 400GE interop issue between customer network router & INFN muxponder
- 2009-05-01 – Infinera President's Award.
- 1991-05-01 – Carleton University Medal for Outstanding Graduate Work at the Ph.D. level. Awarded the best thesis of the year
- 1987-09-01 – Post-Graduate Scholarship from Nortel. Awarded for the duration of the Ph.D. program
- 1991-01-31
- 1987-09-01 – Post-Graduate Scholarship from the Natural Sciences and Engineering Research Council (NSERC) of Canada. Awarded for the duration of the Ph.D. program
- 1990-08-30
- 1978-05-01 – President of India silver medal. Awarded for best academic performance in the graduating batch of the B. Tech program in the Electronics & Communication Engineering (ECE) department

1973-06-01 – Stood fourth in the state of West Bengal in Higher Secondary Examination. The state recognizes the top ten students who appeared for the Higher Secondary Board Examination every year.

Publications

- [1] A. Sadasivarao *et al.*, “Demonstration of Advanced Open WDM Operations and Analytics, Based on an Application-Extensible, Declarative, Data Model Abstracted Instrumentation Platform,” in *2019 Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2019, pp. 1–3.
- [2] M. Anand, R. Subrahmaniam, and R. Valiveti, “POINT: An Intent-Driven Framework for Integrated Packet-Optical In-Band Network Telemetry,” in *2018 IEEE International Conference on Communications (ICC)*, May 2018, pp. 1–6. doi: 10.1109/ICC.2018.8422785.
- [3] M. Anand, R. Subrahmaniam, S. Roy, and R. Valiveti, “Extending Segment Routing into Optical Networks,” in *Optical Fiber Communication Conference*, Optical Society of America, 2017, p. Th1I.3. doi: 10.1364/OFC.2017.Th1I.3.
- [4] M. Anand, R. Subrahmaniam, and R. Valiveti, “Unifying Real-Time Telemetry for Fun and Profit in Packet-Optical Networks,” in *Advanced Photonics 2017 (IPR, NOMA, Sensors, Networks, SPCom, PS)*, Optical Society of America, 2017, p. JTU4A.28. [Online]. Available: <http://www.osapublishing.org/abstract.cfm?URI=Networks-2017-JTU4A.28>
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- [6] F. Dehne, A. Fabri, M. Nassar, A. Rau-Chaplin, and R. Valiveti, “Construction of D-Dimensional Hyperoctrees on a Hypercube Multiprocessor,” *Journal of Parallel and Distributed Computing*, vol. 23, no. 2, pp. 256–261, Nov. 1994, doi: 10.1006/jpdc.1994.1137.
- [7] R. S. Valiveti and B. J. Oommen, “Determining Stochastic Dependence for Normally Distributed Vectors Using the Chi-Squared Metric,” *Pattern Recognition*, vol. 26, no. 6, pp. 975–987, 1993, doi: 10.1016/0031-3203(93)90062-2.
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- [9] R. S. Valiveti and B. J. Oommen, “A Doubly-Linked List Reorganization Heuristic with Stochastic Move-to-End Operations,” in *Proc. of the Twelfth SCCC Intl. Conf. on Computer Science*, Santiago, Chile, Oct. 1992, pp. 249–257.
- [10] F. Dehne, A. Fabri, M. Nassar, A. Rau-Chaplin, and R. S. Valiveti, “Construction of D-Dimensional Hyperoctrees on a Hypercube Multiprocessor,” in *Proc. of 30th Allerton Conference on Communication, Control and Computing*, Sep. 1992, pp. 373–381.
- [11] R. S. Valiveti and B. J. Oommen, “On Using the Chi-Squared Metric for Determining Stochastic Dependence,” *Pattern Recognition*, vol. 25, no. 11, pp. 1389–1400, 1992.
- [12] B. J. Oommen, R. S. Valiveti, and J. R. Zgierski, “An Adaptive Learning Solution to the Keyboard Optimization Problem,” *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 21, no. 6, pp. 1608–1618, Nov. 1991, doi: 10.1109/21.135704.
- [13] R. S. Valiveti and B. J. Oommen, “On Measuring Presortedness in Ensembles of Data Sequences,” in *Proc. of Twenty-ninth Allerton Conf. on Communication, Control, and Computing*, Urbana Champaign, IL, Oct. 1991, pp. 518–523.

- [14] R. S. Valiveti and B. J. Oommen, "The Move-to-Front Heuristic for Non-stationary Query Distributions," in *Proc. of Sixth Intl. Symp. on Computer and Information Sciences (ISCIS VI)*, Antalya, Turkey, Oct. 1991, pp. 105–114.
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- [16] R. S. Valiveti and B. J. Oommen, "A Syntactic-Statistical Pattern Recognition Approach to Distinguishing between Encryption Keys," in *Proc. of IASTED Intl. Symp. on Artificial Intelligence Applications and Neural Networks*, Zurich, Switzerland, Jul. 1991, pp. 122–124.
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Patents

- [1] R. Valiveti, R. Rao, V. Narippatta, S. Sharfuddin, and P. Kandappan, "Providing Access to Client Overhead While Transmitting the Client Signal over Multiple Optical Sub-Carriers," 12224800B2, Feb. 11, 2025, <https://patents.google.com/patent/US12224800B2/en>.
- [2] R. Rao, R. Pratapa, R. Srinivasan, A. Khunjidhapatham, and R. Valiveti, "Sd-Fec Defect Propagation across Regeneration Nodes in Support of Single and Multi-Carrier Super Channel Protection in Roadm Networks," 11838048B2, Jul. 22, 2021, <https://patents.google.com/patent/US11838048B2/en>.

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- [14] E. E. Sprague *et al.*, "Providing Access to Client Overhead While Transparently Transmitting the Client Signal," 8446906B2, May 21, 2013, <https://patents.google.com/patent/US8446906B2/>.
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- [16] M. H. Hui, R. S. Valiveti, H. Patel, J. A. Lamont, and Y. B. Shah, "Remote Interconnection of Local Area Networks," 4901312A, Feb. 13, 1990, <https://patents.google.com/patent/US4901312A/en?inventor=radhakrishna+valiveti&page=1>.
- [17] M. Anand, R. Subrahmaniam, and R. Valiveti, "Elastic Timestamping," 20190013954A1, <https://patents.google.com/patent/US20190013954A1/>.
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- [19] R. Rao, R. Valiveti, M. Plantaire, and S. Sharfuddin, "Method and Apparatus for Coherent Optical Module Encryption and Decryption," 20260128pending.

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